

Data Science Across Domains

From Methods and Models to Business Applications

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Table of contents

Welcome	1
Preface	3
About	5
How to Read This Portfolio	7
I Quantitative Finance	9
Overview	11
1 CAPM and Fama–French Models	13
1.1 Problem Definition	13
1.2 Modeling Approach	14
1.3 Methodology and Implementation	18
1.3.1 Data Collection and Preprocessing	18
1.3.2 CAPM Implementation	18
1.3.3 Fama–French Three-Factor Model Implementation	23
1.4 Results, Discussion, and Limitations	26
1.4.1 Comparative Analysis	27
1.4.2 Model Limitations	28
1.5 Technical Appendix (Optional)	29
1.5.1 Simple Linear Regression and the CAPM Model	30
1.5.1.1 Ordinary Least Squares Estimation	30
1.5.1.2 Residual Analysis and Model Assumptions	32

1.5.1.3	Confidence Intervals and Standard Errors . . .	33
1.5.1.4	Hypothesis Testing and Statistical Significance .	34
1.5.1.5	Analysis of Variance (ANOVA)	35
1.5.1.6	Coefficient of Determination (R^2)	38
1.5.2	Multiple Linear Regression and the Fama-French Model .	39
1.5.2.1	Ordinary Least Squares Estimation	40
1.5.2.2	Statistical Interpretation of the Coefficients . .	41
1.5.2.3	Global Hypothesis Testing and Model Signifi- cance	42
1.5.2.4	Limitations of Multiple Linear Regression . . .	43
2	Unsupervised Learning for Asset Allocation	45
2.1	Problem Definition	45
2.2	Modeling Approach	45
2.3	Methodology and Implementation	46
2.4	Results, Discussion, and Limitations	47
2.5	Technical Appendix (Optional)	47
II	Retail Analytics	49
	Overview	51
3	Credit Card Fraud Detection	53
3.1	Problem Definition	53
3.2	Modeling Approach	53
3.3	Methodology and Implementation	54
3.4	Results, Discussion, and Limitations	55
3.5	Technical Appendix (Optional)	55
4	Customer Lifetime Value Prediction	57
4.1	Problem Definition	57
4.2	Modeling Approach	57
4.3	Methodology and Implementation	58
4.4	Results, Discussion, and Limitations	59
4.5	Technical Appendix (Optional)	59
5	Marketing Campaign Optimization	61
5.1	Problem Definition	61

5.2	Modeling Approach	61
5.3	Methodology and Implementation	62
5.4	Results, Discussion, and Limitations	63
5.5	Technical Appendix (Optional)	63
6	RFM Segmentation with K-Means	65
6.1	Problem Definition	65
6.2	Modeling Approach	65
6.3	Methodology and Implementation	66
6.4	Results, Discussion, and Limitations	67
6.5	Technical Appendix (Optional)	67
7	Customer Sentiment and Topic Analysis	69
7.1	Problem Definition	69
7.2	Modeling Approach	69
7.3	Methodology and Implementation	70
7.4	Results, Discussion, and Limitations	71
7.5	Technical Appendix (Optional)	71
III	Biological Systems	73
	Overview	75
8	Microbial Population Dynamics (Lotka–Volterra Model)	77
8.1	Problem Definition	77
8.2	Modeling Approach	77
8.3	Methodology and Implementation	78
8.4	Results, Discussion, and Limitations	79
8.5	Technical Appendix (Optional)	79
IV	Energy and Industrial Systems	81
	Overview	83
9	Geochemical Fingerprint Deconvolution	85
9.1	Problem Definition	85
9.2	Modeling Approach	85

9.3	Methodology and Implementation	86
9.4	Results, Discussion, and Limitations	87
9.5	Technical Appendix (Optional)	87
10	Pore–Throat Detection in Porous Media	89
10.1	Problem Definition	89
10.2	Modeling Approach	89
10.3	Methodology and Implementation	90
10.4	Results, Discussion, and Limitations	91
10.5	Technical Appendix (Optional)	91
11	Optimal Hedging of Industrial Gas Demand	93
11.1	Problem Definition	93
11.2	Modeling Approach	93
11.3	Methodology and Implementation	94
11.4	Results, Discussion, and Limitations	95
11.5	Technical Appendix (Optional)	95
12	Knowledge Extraction from Technical Documents with NLP and LLMs	97
12.1	Problem Definition	97
12.2	Modeling Approach	97
12.3	Methodology and Implementation	98
12.4	Results, Discussion, and Limitations	99
12.5	Technical Appendix (Optional)	99
V	Climate and Environmental Systems	101
	Overview	103
13	Climate Forecasting with Time Series	105
13.1	Problem Definition	105
13.2	Modeling Approach	105
13.3	Methodology and Implementation	106
13.4	Results, Discussion, and Limitations	107
13.5	Technical Appendix (Optional)	107

VI Writings & Essays	109
Overview	111
14 Data Science in the Age of Hype: Why the scientific method still matters	113
14.1 The Rise of Data Science	113
14.2 The Nature of Science and the Scientific Method	116
14.3 The Scientific Mindset of a Data Scientist	119
14.4 The Fundamental Constituents of Data Science	122
14.5 Data Science as a Strategic Value Engine	122
14.6 Why the Scientific Method Still Matters	124
15 Conformal Geometry and Electrostatics in Three Dimensions	125
15.1 Laplace's equation	125
15.2 Conformal Mappings	125
15.3 Problem Definition	125
15.4 Modeling Approach	126
15.5 Methodology and Implementation	126
15.6 Results, Discussion, and Limitations	127
15.7 Technical Appendix (Optional)	127
References	131

Welcome

Welcome to my portfolio:

Data Science Across Domains

From Methods and Models to Business Applications

This portfolio presents a curated collection of applied projects developed across different domains. Rather than a traditional theoretical text, it is organized as a series of case studies illustrating how quantitative methods, machine learning, and computational modeling can be used to address real-world problems.

Each chapter focuses on a specific project. Problems are clearly defined, modeling assumptions are made explicit, and solutions are developed through a combination of statistical reasoning, computational experimentation, and quantitative modeling. Results are analyzed and interpreted within the context of the problem being addressed.

The emphasis throughout the portfolio is on applied problem solving. Mathematical concepts and technical details are introduced only when they help clarify modeling decisions, support methodological choices, or explain the behavior of the proposed approaches.

Every project presented here corresponds to a fully implemented workflow. The associated computational artifacts—written in Python and R—are available alongside the text, allowing readers to explore the code, reproduce experiments, and examine the implementation details supporting each solution.

This portfolio reflects an approach to applied data science and quantitative modeling that combines structured reasoning, careful experimentation, and practical implementation.

Many of the projects presented here also draw on my background in physics

and computational modeling, where mathematical reasoning and computational experimentation naturally complement data-driven approaches to problem solving.

Each chapter can be read independently as a standalone project, while together they illustrate a consistent approach to developing technically sound and practically meaningful solutions across different domains.

Preface

This portfolio reflects an ongoing effort to organize my work in applied data science and quantitative modeling into a coherent and practice-oriented body of projects. Rather than presenting isolated experiments or purely theoretical discussions, it documents complete workflows in which problem formulation, modeling choices, implementation, and evaluation are treated as parts of the same analytical process.

Structuring this material in book form mirrors how I approach quantitative and computational problems in professional and research environments. In practice, successful applications rarely depend on sophisticated methods alone. They usually emerge from a clear definition of the problem, explicit modeling assumptions, careful experimentation, and thoughtful interpretation of results within the context of the domain.

For this reason, the chapters are organized around concrete case studies. Each project begins with a well-defined problem and progresses through methodological design, implementation, evaluation, and discussion of limitations. Concepts from statistics, optimization, probability, numerical methods, and machine learning appear throughout the text, but only when they are necessary to support modeling decisions or clarify the behavior of the proposed approaches.

An equally important aspect of this portfolio is its deliberate scope. It does not attempt to catalogue every possible technique or follow trends for their own sake. Instead, it focuses on methods that I have implemented, tested, and evaluated in real problem settings. The emphasis is placed on depth, clarity, and practical relevance rather than on breadth.

All projects presented here correspond to fully implemented solutions. Computational artifacts written in Python and R accompany the text and are referenced

throughout, allowing readers to explore the implementation and reproduce the results when appropriate.

Ultimately, this portfolio serves both as a record of completed work and as a foundation for continued development. The projects documented here represent practical steps in an ongoing process of learning, experimentation, and refinement. My intention is that the structure of this work reflects not only technical capability, but also a disciplined and pragmatic approach to quantitative modeling and computational problem solving in real-world contexts.

About

I am a physicist and data scientist with a background in applied mathematics, machine learning, and computational modeling. My work focuses on applying quantitative methods to real-world problems, combining mathematical reasoning with practical implementation.

My academic training in physics has strongly shaped the way I approach quantitative modeling and data-driven problem solving. My research included work on **quantum optimal control theory**, with particular emphasis on algorithmic design based on *shortcuts to adiabaticity* and the study of computational complexity in physical systems. This background continues to influence my modeling philosophy through an emphasis on clear problem formulation, explicit assumptions, and careful evaluation of model behavior.

Professionally, I have applied machine learning, statistical modeling, and optimization techniques across data-intensive industries. Much of my work has been developed within the **energy sector**, where I have contributed to quantitative modeling, decision-support systems, and optimization frameworks for gas trading and operational planning.

In the retail sector, I have worked on projects involving customer segmentation, churn prediction, and recommendation systems, balancing quantitative modeling with practical constraints such as data quality, interpretability, and deployment considerations.

More recently, within Oil & Gas research and development environments, my work has focused on integrating physical modeling, statistical inference, and machine learning to address complex industrial problems. These projects include predictive modeling for pipeline corrosion, production optimization, and simulation-based approaches for decision-making under uncertainty.

Across both academic and industrial settings, my perspective remains consistent: computational and statistical methods are most effective when supported by disciplined modeling, transparent assumptions, and careful validation.

Outside my professional work, I enjoy playing the bass guitar and maintain a longstanding interest in theology.

How to Read This Portfolio

This portfolio is organized around applied projects rather than purely theoretical chapters. Each chapter presents a self-contained case study illustrating how quantitative methods, machine learning, and computational modeling can be used to address real-world problems across different domains.

Projects can be read independently. Readers interested in a particular topic may navigate directly to the corresponding chapter without following the entire sequence.

Most projects follow a common structure:

1. **Problem Definition**
Description of the problem, its context, and its practical relevance within the domain.
2. **Modeling Approach**
Presentation of the conceptual framework, modeling assumptions, and methodological reasoning underlying the proposed solution.
3. **Methodology and Implementation**
Description of the data, computational workflow, experimental design, and implementation details used throughout the project.
4. **Results, Discussion, and Limitations**
Presentation and interpretation of results together with an analysis of model behavior, assumptions, limitations, and possible extensions.
5. **Technical Appendix (Optional)**
Supplementary technical material, including mathematical derivations, methodological notes, and additional computational details when relevant to the project.

Theoretical concepts and mathematical details are introduced only when they help clarify modeling decisions, support methodological choices, or explain the behavior of the proposed approaches. The emphasis throughout the portfolio remains on applied problem solving, structured reasoning, and practical implementation rather than on purely theoretical discussion.

Each chapter is accompanied by computational artifacts implemented in Python and R, allowing readers to explore the implementation details, reproduce experiments, and examine complementary analyses when appropriate.

The material presented here combines quantitative reasoning, statistical modeling, machine learning methods, and computational experimentation, with a consistent focus on developing solutions that are both technically sound and practically meaningful.

Part I

Quantitative Finance

Overview

This part covers problems related to quantitative finance, including the modeling of financial assets, portfolio construction, and risk analysis under uncertainty.

The scope of this domain includes methods for analyzing market data, evaluating financial instruments, and supporting decision-making where risk and return must be jointly considered.

Chapter 1

CAPM and Fama–French Models

This project studies how classical and multi-factor asset pricing models explain the relationship between systematic risk and expected return. Using historical market data, the analysis evaluates the assumptions, explanatory power, and empirical limitations of the CAPM and Fama–French frameworks through regression-based factor modeling.

1.1 Problem Definition

This project evaluates how asset pricing models—from the traditional CAPM to the multi-factor extensions of Fama and French—formalise the relationship between risk and expected return. In practice, an investor faced with a price signal must determine whether the potential return justifies the associated risk. Doing this effectively means discounting uncertain future cash flows back to present value, which requires a reliable discount rate. The core question is straightforward: what minimum return should be demanded from an asset given its specific risk profile?

Getting this right is critical for asset valuation, portfolio construction, and fund performance evaluation. Without a structured framework, investment decisions often default to guesswork or oversimplified rules of thumb—such as chasing past returns or assuming that any form of risk automatically deserves a premium. Modern financial theory offers a more rigorous perspective: the market does not reward idiosyncratic or stock-specific risk, since it can be diversified away.

Only systematic risk warrants a higher expected return. This principle is what separates the CAPM from more naive approaches and serves as the foundation for multi-factor models.

Key decisions—ranging from efficient capital allocation to identifying market mispricings—rely heavily on these models. If an asset offers an expected return above what its systematic risk predicts, it may indicate undervaluation. However, the real-world challenge is methodological: true expected returns are unobservable. Relying on historical data introduces noise, sample-period sensitivity, and specification errors. Furthermore, the CAPM compresses all systematic risk into a single factor (the market portfolio), a simplification that empirical evidence has repeatedly challenged.

The Fama–French models attempt to address this limitation by capturing additional dimensions of systematic risk, including size, value, profitability, and investment patterns. From a data science perspective, this project explores how these models are constructed, tests their underlying assumptions, and evaluates how well they fit empirical data. The final output is a comparative analysis of the models’ explanatory power and practical limitations when confronted with real-world financial data that rarely behaves as textbook theory predicts.

1.2 Modeling Approach

To approach the problem of asset valuation, this project is grounded on two fundamental principles of finance. The first is the time value of money: a dollar today is worth more than a dollar tomorrow because it can be invested immediately and start generating returns. The second is risk aversion: an uncertain cash flow is only attractive if it offers additional compensation relative to a risk-free alternative. Together, these principles reflect the two central variables underlying any financial valuation problem: time and risk.

Under this framework, the valuation of an asset can be understood as the present value of an uncertain future cash flow:

$$PV = \left(\frac{1}{1 + r_t} \right)^t \mathbb{E}[X_t] \quad (1.1)$$

Here, PV represents the expected present value of a future cash flow X_t received

at time t , while r_t denotes the discount rate associated with risk and the opportunity cost of capital. The fundamental challenge for the investor is that this discount rate is not known in advance. In other words, the key question is how much return should be required to compensate for the level of risk being taken. This problem provides the conceptual motivation for the Capital Asset Pricing Model (CAPM) and its multifactor extensions.

Since future returns are uncertain, this project models them as random variables. A practical way to characterize them is through their expected value, representing the anticipated average return, and their volatility, which measures the dispersion of outcomes around that average. This mean-variance framework, originally developed by Harry Markowitz in modern portfolio theory, shows that total portfolio risk can be reduced by combining assets whose returns do not move exactly in the same way [1]. The intuition is straightforward: through diversification, part of the individual fluctuations of each asset tends to offset one another.

However, diversification has an observable practical limit. As portfolios are constructed with an increasing number of assets, total volatility initially falls rapidly but eventually stabilizes around a lower bound. As illustrated in Figure 1.1, this behavior reveals the existence of two distinct components of risk. On the one hand, idiosyncratic or diversifiable risk, which is associated with firm-specific events and can largely be eliminated through diversification. On the other hand, systematic risk, which is linked to broader macroeconomic and market-wide movements and therefore persists even in highly diversified portfolios.

This distinction has a fundamental implication for asset pricing. In a competitive market, rational investors should not expect compensation for taking risks that can be eliminated at virtually no cost through diversification. As a result, the market only rewards exposure to systematic risk. This idea forms the conceptual foundation of the CAPM, which states that the expected risk premium of an asset should be proportional to the expected risk premium of the market [2] [3]:

$$\mathbb{E}[r_i] - r_f = \beta_i (\mathbb{E}[r_m] - r_f) \quad (1.2)$$

In this expression, $\mathbb{E}[r_i]$ represents the expected return of asset i , r_f is the risk-free rate, and $\mathbb{E}[r_m]$ corresponds to the expected return of the market portfolio. The parameter β_i measures the sensitivity of the asset to market fluctuations and summarizes its exposure to systematic risk.

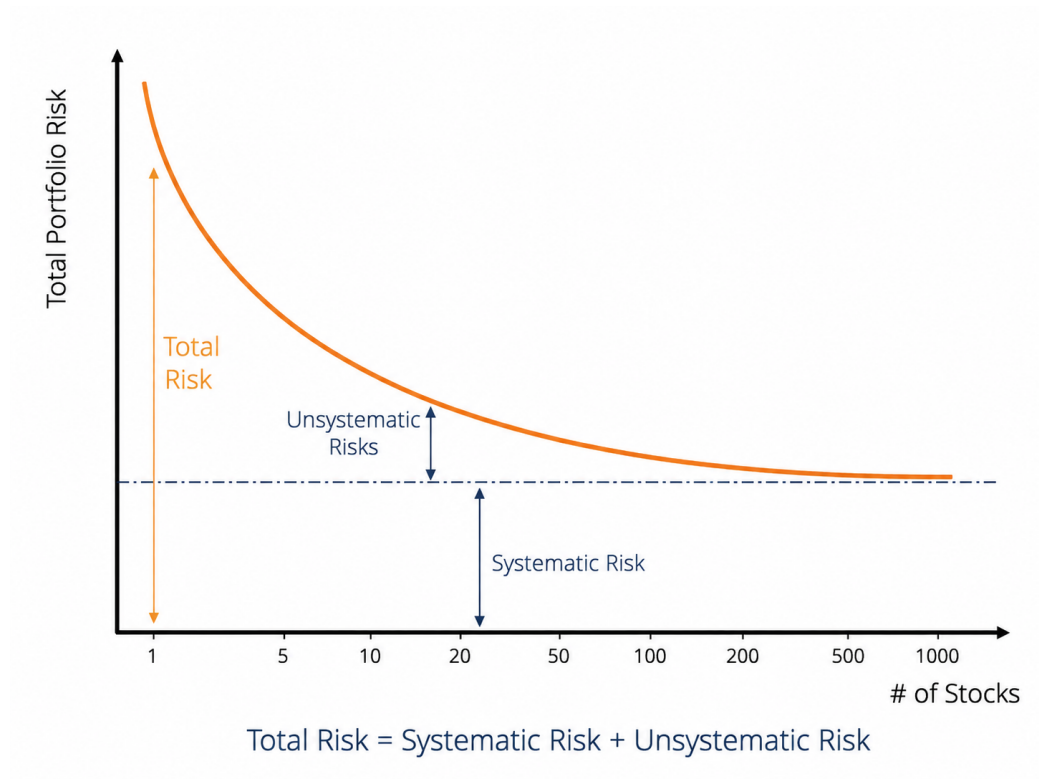


Figure 1.1: Decomposition of total risk into systematic and diversifiable risk as assets are progressively added to a portfolio.

From a modeling perspective, the estimation of beta can be interpreted as a simple linear regression between asset returns and market returns [4]. In this project, the CAPM is implemented precisely under this framework, using simple linear regression to estimate the market exposure of different assets and analyze the relationship between systematic risk and expected return.

Despite its conceptual simplicity and historical importance, the CAPM assumes that the entire structure of systematic risk can be summarized by a single factor: the market itself. Empirical evidence, however, suggests that certain firm characteristics — such as company size or the relationship between book value and market value — capture persistent return patterns that the CAPM does not fully explain. These observations motivated the development of the multifactor models proposed by Fama and French [5] [6].

In this project, these extensions are implemented using multiple linear regression. The three-factor model can be expressed as:

$$\mathbb{E}[r_i] - r_f = \alpha + \beta_{mkt}(\mathbb{E}[r_m] - r_f) + \beta_{smb} \cdot \text{SMB} + \beta_{hml} \cdot \text{HML} + \varepsilon \quad (1.3)$$

where SMB (*Small Minus Big*) measures the excess return of small-cap stocks relative to large-cap stocks, while HML (*High Minus Low*) captures the return differential between value and growth stocks [7] [8].

From a modeling perspective, the transition from the CAPM to the Fama–French three-factor model can be interpreted as moving from a simple linear regression to a multiple linear regression framework, incorporating additional explanatory variables to capture dimensions of systematic risk that a single-factor model cannot fully represent.

The main objective of the project is not simply to compare statistical goodness-of-fit metrics, but to understand how different economic factors contribute to the generation of returns in financial markets and to evaluate how effectively models of different complexity levels represent that underlying process.

1.3 Methodology and Implementation

1.3.1 Data Collection and Preprocessing

The analysis uses daily adjusted closing prices obtained from Yahoo Finance for both Apple Inc. (AAPL) and the S&P 500 index (^GSPC). The S&P 500 is used as a proxy for the market portfolio within the CAPM framework.

The selected period spans from January 2020 to December 2024, providing a multi-year sample that includes different market conditions and volatility regimes. Adjusted prices are used in order to account for corporate actions such as stock splits and dividend adjustments when available.

After downloading the data, both time series are merged into a single dataset and aligned by trading date. Missing observations are removed prior to return calculation.

The price series are initially inspected to verify data consistency and to obtain a preliminary view of the market dynamics during the analyzed period.

Daily simple returns are then computed from the adjusted closing prices. These returns constitute the main variables used throughout the regression analysis.

Before estimating the model, the relationship between asset and market returns is visually explored through a scatter plot. As shown in Figure 1.3, the data already suggest a positive linear relationship between Apple returns and market returns, supporting the use of a linear regression framework for the CAPM estimation.

1.3.2 CAPM Implementation

Following the theoretical formulation introduced in Equation 1.2, the CAPM is implemented through a simple linear regression in which the daily returns of Apple Inc. are modeled as a function of the daily returns of the market portfolio.

For simplicity, the implementation presented here uses raw daily returns without explicitly incorporating a risk-free rate. Under this approximation, the model estimates the linear sensitivity of the selected asset relative to market movements.

The estimated regression coefficients are summarized below.



Figure 1.2: Adjusted closing prices for Apple Inc. and the S&P 500 index between 2020 and 2024. The figure is mainly used as a preliminary inspection of the downloaded financial time series before return calculation.



Figure 1.3: Scatter plot of daily Apple returns against S&P 500 daily returns. The figure provides an exploratory visualization of the positive relationship between the asset and market returns prior to the CAPM estimation.

Table 1.1: Estimated CAPM regression coefficients for Apple Inc. using the S&P 500 as the market proxy.

Term	Estimate	Std. Error	t-statistic	p-value
(Intercept)	0.000526	0.000346	1.5226	0.128105
market	1.173257	0.025682	45.6847	0.000000

The estimated market beta is approximately $\beta = 1.173257$, indicating that Apple exhibited higher sensitivity to market fluctuations than the S&P 500 during the analyzed period. Under the CAPM interpretation, this coefficient suggests that, on average, Apple returns moved approximately 17% more aggressively than the market portfolio in response to market variations.

Additional model diagnostics are reported below.

Table 1.2: Diagnostic statistics associated with the CAPM linear regression model.

Metric	Value
R-squared	0.6247
Adjusted R-squared	0.6244
F-statistic	2087.0916
Observations	1256

The estimated beta coefficient is also statistically significant, with a large associated t -statistic and a near-zero p -value. Similarly, the overall regression presents a large F -statistic, supporting the existence of a statistically meaningful linear relationship between Apple returns and market returns during the analyzed period.

The estimated coefficient of determination of approximately $R^2 = 0.625$ indicates that the market factor alone explains a substantial fraction of the observed variability in Apple daily returns during the selected period. Nevertheless, a significant portion of the variability remains unexplained, motivating the consideration of multifactor extensions beyond the single-factor CAPM specification.

The fitted linear relationship between asset and market returns is illustrated in Figure 1.4.

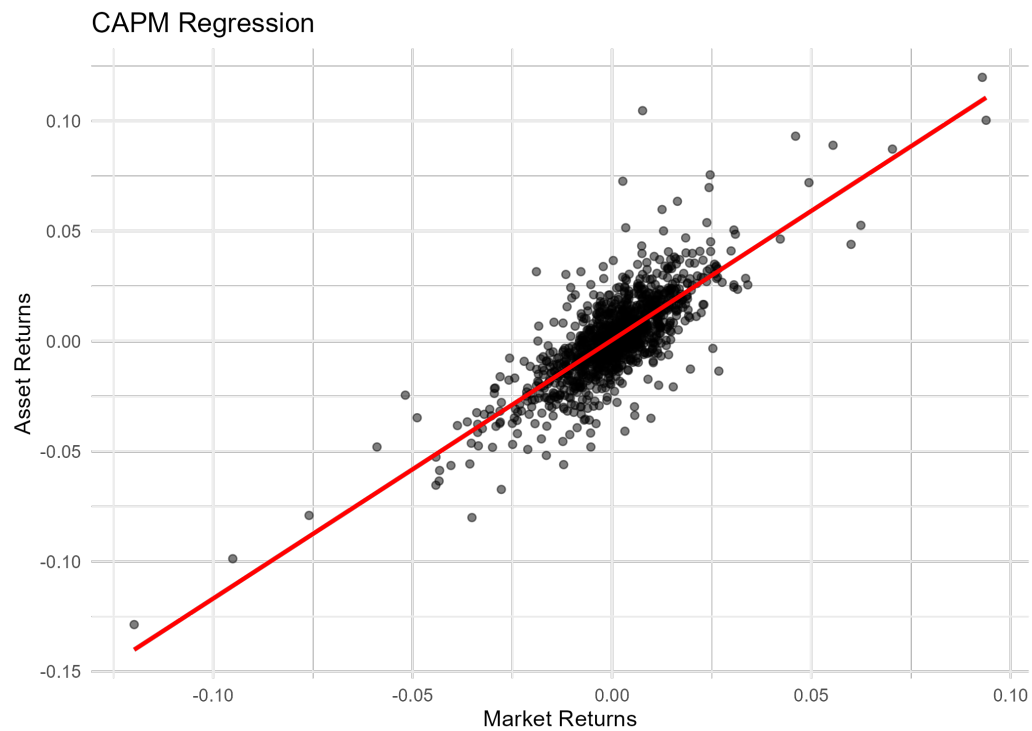


Figure 1.4: CAPM linear regression between Apple daily returns and S&P 500 daily returns. The red line represents the fitted linear relationship estimated through ordinary least squares regression.

Additional implementation details are available in the companion [Quarto notebook](#) here. The corresponding Python notebooks associated with this project are also available in the repository .

1.3.3 Fama–French Three-Factor Model Implementation

To extend the single-factor CAPM specification, the Fama–French three-factor model introduced in Equation 1.3 was estimated using ordinary least squares (OLS) regression. In addition to the market excess return factor, the model incorporates two additional systematic risk factors associated with firm size (SMB) and relative valuation (HML).

The implementation combines daily adjusted closing prices for Apple Inc. (AAPL) obtained from Yahoo Finance with the daily Fama–French factor dataset developed by Kenneth French. The final modeling dataset contains 1,256 daily observations covering the period from January 2020 to December 2024. The average daily excess return of Apple over the sample period was approximately 0.11%, with a daily standard deviation close to 2.0%.

Additional implementation details are available in the companion [Quarto notebook](#) here. The corresponding Python notebooks associated with this project are also available in the repository .

Figure 1.5 displays the behavior of the three explanatory factors over time. The series exhibit substantial short-term variability, particularly during periods of market stress such as the early stages of the COVID-19 pandemic in 2020. The SMB factor shows the highest volatility among the three factors during this period, reflecting the stronger sensitivity of small-cap firms to abrupt changes in market conditions.

The estimated regression coefficients are summarized below.

Table 1.3: Estimated Fama–French three-factor regression coefficients.

Term	Estimate	Std. Error	t-statistic	p-value
Alpha	0.000455	0.000320	1.421686	0.155366
Market Factor	1.162676	0.024001	48.442699	0.000000
Size Factor	-0.329641	0.042313	-7.790601	0.000000
Value Factor	-0.370591	0.027910	-13.278185	0.000000

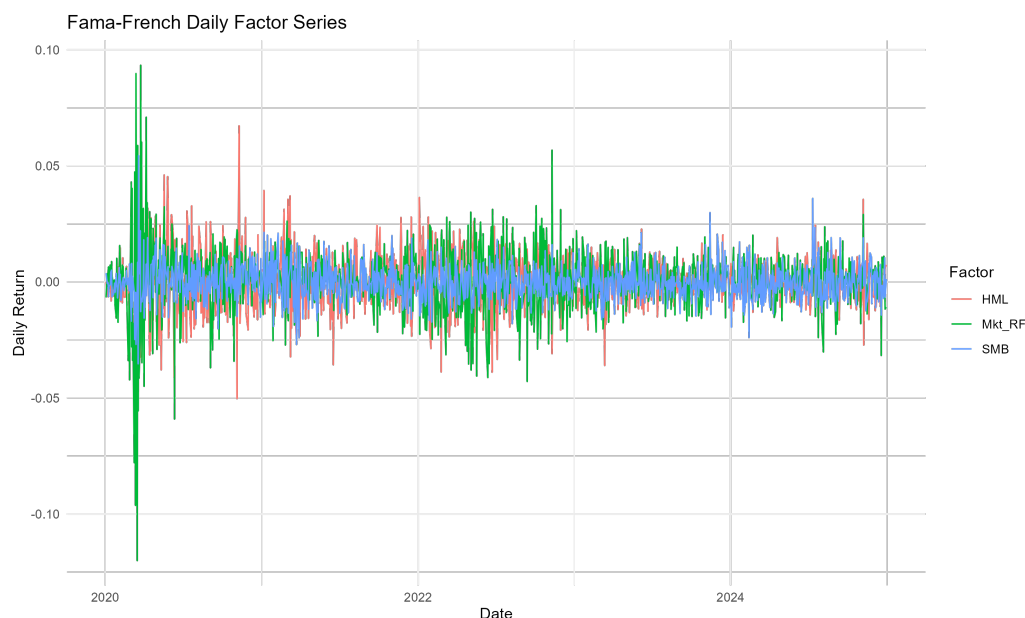


Figure 1.5: Fama–French daily factor series.

The estimated market beta is approximately $\beta_{mkt} = 1.163$, indicating that Apple exhibits a stronger sensitivity to systematic market movements than the market portfolio itself. The coefficients associated with the size and value factors are both negative and statistically significant. The negative loading on SMB suggests that Apple behaves more similarly to large-cap firms than to small-cap firms, which is consistent with its market capitalization and dominant position within the technology sector. Likewise, the negative HML exposure indicates characteristics more closely aligned with growth-oriented firms rather than value stocks.

The estimated intercept term ($\alpha \approx 0.00046$) is positive but not statistically significant at conventional significance levels ($p \approx 0.155$), suggesting that, after accounting for the three systematic risk factors, the model does not provide strong evidence of abnormal excess returns during the analyzed period.

The overall regression fit improved relative to the single-factor CAPM specification. The coefficient of determination increased to approximately $R^2 = 0.678$, indicating that the three-factor model explains nearly 68% of the variability observed in Apple excess returns. The F-statistic associated with the regression is also highly significant, supporting the joint explanatory power of the selected

factors.

Table 1.4: Diagnostic statistics associated with the Fama-French three-factor regression model.

Metric	Value
R-squared	0.677865
Adjusted R-squared	0.677093
Sigma	0.011345
F-statistic	878.191163
p-value	0.000000
Observations	1256

Figure 1.6 compares the observed excess returns against the fitted values generated by the regression model. The fitted series captures a substantial fraction of the short-term dynamics of Apple returns, although large residual fluctuations remain during periods of elevated volatility.



Figure 1.6: Observed and fitted excess returns under the Fama–French three-factor specification.

The residual series displayed in Figure 1.7 remains approximately centered around zero throughout the analyzed period, without obvious long-term structural drift. However, clusters of higher residual volatility are still visible during stressed market conditions, indicating that a portion of the variability remains unexplained even after incorporating the additional systematic factors.

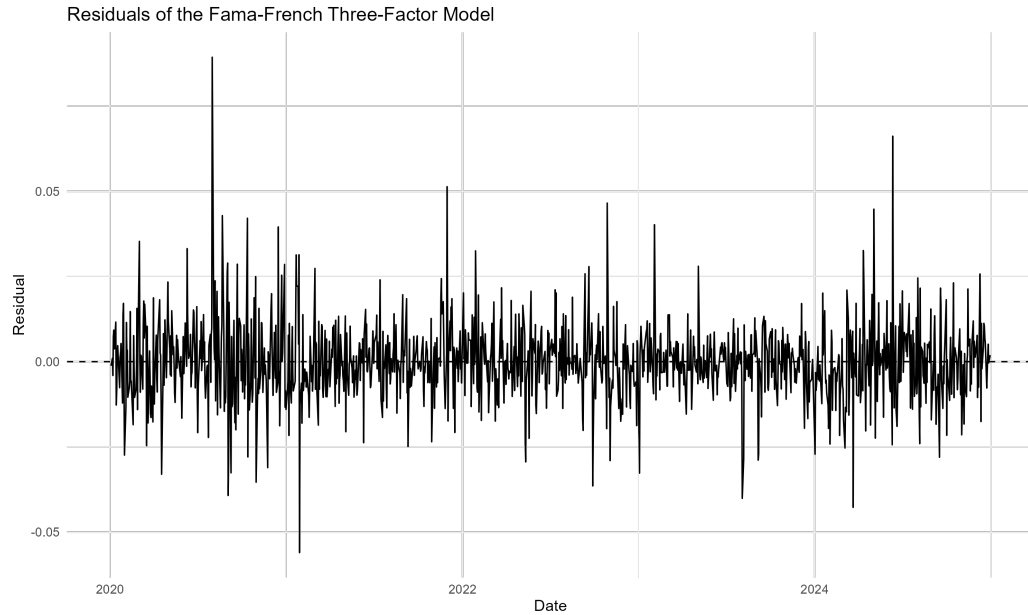


Figure 1.7: Residual series of the Fama–French three-factor model.

Finally, the residual distribution shown in Figure 1.8 exhibits a roughly symmetric shape centered near zero, although moderate tail behavior remains visible. This behavior is consistent with the presence of occasional large market movements that are not fully captured by the linear factor structure.

1.4 Results, Discussion, and Limitations

The empirical results obtained from both the CAPM and the Fama–French three-factor model provide a useful comparison between single-factor and multifactor approaches to asset pricing. Although both models capture an important fraction of the variability observed in Apple excess returns, the inclusion of additional systematic risk factors in the Fama–French specification produces a measurable improvement in explanatory performance.

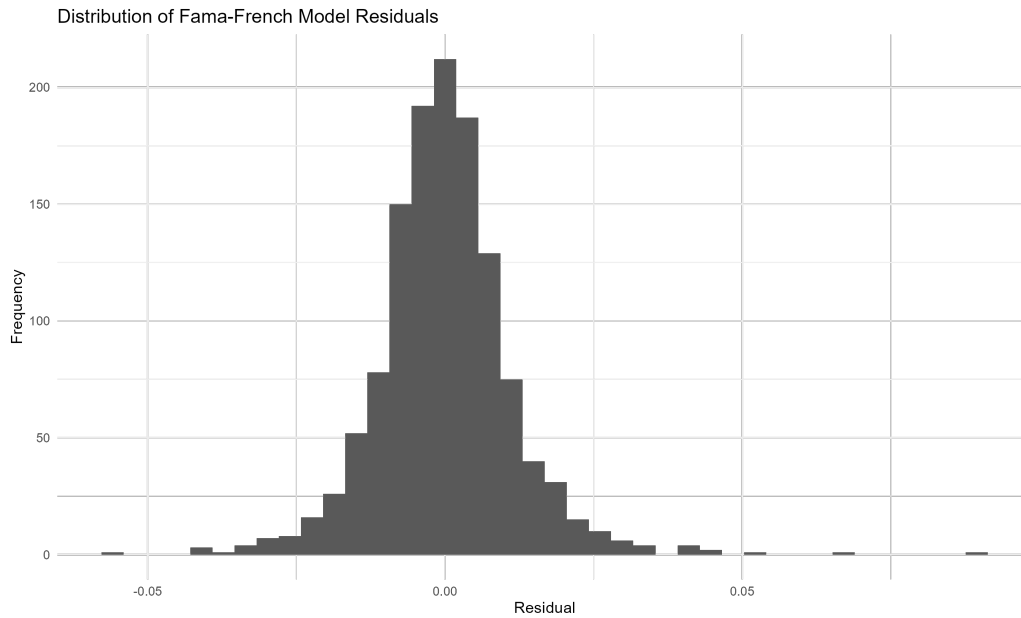


Figure 1.8: Distribution of residuals from the Fama–French three-factor model.

1.4.1 Comparative Analysis

The CAPM estimation produced a market beta close to $\beta \approx 1.17$, indicating that Apple exhibits a level of systematic risk slightly higher than that of the overall market portfolio. The market factor alone explained approximately 62.5% of the variability in Apple excess returns over the analyzed period, resulting in an estimated coefficient of determination of approximately $R^2 = 0.625$.

After incorporating the size (SMB) and value (HML) factors, the explanatory power of the regression increased to approximately $R^2 = 0.678$. Although the increase in explanatory performance is moderate, the multifactor specification captures an additional fraction of return variability that remains unexplained under the single-factor CAPM framework.

The estimated market beta remained relatively stable across both models, suggesting that market risk continues to represent the dominant systematic component driving Apple returns. However, the additional factor loadings provide further insight into the firm's structural characteristics. The negative exposure to the size factor indicates that Apple behaves more similarly to large-cap firms than to smaller companies, while the negative loading associated with the value

factor reflects characteristics commonly associated with growth-oriented firms.

From a statistical perspective, both models produced highly significant overall regressions according to their respective F-statistics. Nevertheless, the Fama–French specification generated lower residual variability and a modest improvement in fitted return dynamics, particularly during periods of elevated market volatility.

The comparison between observed and fitted excess returns also suggests that the multifactor specification is better able to reproduce short-term return fluctuations than the CAPM alone. However, significant residual movements remain visible in both models, indicating that a substantial fraction of the variability in daily equity returns cannot be fully captured through linear factor structures alone.

Overall, the empirical results support the interpretation that the Fama–French three-factor model provides a richer and more flexible representation of systematic risk exposures relative to the standard CAPM specification. At the same time, the relatively modest improvement in explanatory power also highlights the dominant role played by market-wide risk in large-cap technology firms such as Apple.

1.4.2 Model Limitations

Despite their widespread use in financial economics, both the CAPM and the Fama–French three-factor model rely on simplifying assumptions that limit their ability to fully explain observed market behavior.

First, both models assume linear relationships between excess returns and systematic risk factors. In practice, financial markets frequently exhibit nonlinear dynamics, volatility clustering, structural breaks, and regime-dependent behavior that cannot be fully represented within a static linear regression framework.

Second, the analysis relies on historical daily observations and therefore assumes that the estimated factor sensitivities remain relatively stable throughout the selected time horizon. However, the relationship between individual assets and systematic risk factors may evolve over time as market conditions, macroeconomic environments, or firm-specific characteristics change.

An additional limitation arises from the selection of explanatory factors themselves. Although the Fama–French specification extends the CAPM by incorpo-

rating size and value effects, many additional sources of systematic risk have been proposed in the literature, including profitability, investment, momentum, liquidity, and volatility-related factors. Consequently, the three-factor model should not be interpreted as a complete description of equity return dynamics.

The analysis also focuses exclusively on a single large-cap technology company. While Apple represents an interesting case study due to its market relevance and liquidity, the estimated coefficients and model diagnostics cannot be directly generalized to other sectors, asset classes, or market environments without additional empirical validation.

Finally, both models are estimated under the assumption that historical relationships provide meaningful information regarding future risk-return behavior. As in most empirical financial applications, this assumption may fail under changing market conditions, particularly during periods of financial stress or structural economic transitions.

Despite these limitations, the CAPM and Fama–French frameworks continue to provide valuable benchmark models for understanding systematic risk exposures, comparing relative asset sensitivities, and constructing interpretable factor-based representations of financial returns.

1.5 Technical Appendix (Optional)

This appendix provides a compact technical background for the regression models used in this project. The main body of the chapter focuses on the financial interpretation of the CAPM and Fama–French models, while this appendix summarizes the statistical structure underlying their empirical estimation.

The CAPM specification can be interpreted as a simple linear regression model, where the excess return of an asset is explained by the excess return of the market portfolio. In contrast, the Fama–French three-factor model extends this idea to a multiple linear regression setting, where asset excess returns are explained by a set of risk factors rather than by the market factor alone.

The purpose of this appendix is not to provide a complete treatment of linear regression or statistical learning. Instead, it collects the main concepts needed to understand how the models were estimated, how the coefficients should be interpreted, and how goodness-of-fit and statistical significance measures relate

to the empirical results discussed in the project. The presentation follows standard treatments of linear regression and statistical learning [9] [10] [11] [12] [13] [14].

1.5.1 Simple Linear Regression and the CAPM Model

As discussed previously in Equation 1.2, the CAPM establishes a linear relationship between the expected excess return of an asset and the expected excess return of the market portfolio. From a statistical perspective, this relationship can be interpreted as a simple linear regression model of the form

$$y_i = \theta_0 + \theta_1 x_i + \varepsilon_i \quad (1.4)$$

where θ_0 represents the intercept, θ_1 the slope coefficient, and ε_i an unobservable random error associated with observation i .

Under this interpretation, the empirical CAPM specification can be written as

$$r_i - r_f = \alpha + \beta(r_m - r_f) + \varepsilon \quad (1.5)$$

where the excess return of the asset plays the role of the response variable and the excess return of the market portfolio acts as the predictor variable. In this setting, α corresponds to the intercept term, while β measures the sensitivity of the asset return to market movements.

From the perspective of the CAPM, the intercept term α is expected to be equal to zero, implying that expected excess returns are fully explained by exposure to market risk through the coefficient β . Consequently, testing whether α differs significantly from zero provides a practical way to evaluate the empirical consistency of the model.

1.5.1.1 Ordinary Least Squares Estimation

The parameters of the regression model are estimated by minimizing the discrepancy between the observed data and the fitted regression line. In ordinary least squares (OLS), this discrepancy is quantified through the residual sum of squares (RSS), defined as

$$\text{RSS} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 = \sum_{i=1}^N (y_i - \theta_0 - \theta_1 x_i)^2 \quad (1.6)$$

where each residual represents the difference between an observed value and the value predicted by the model.

Figure 1.9 illustrates the geometric interpretation of the residuals in simple linear regression. The ordinary least squares estimator selects the parameters that minimize the squared residual distances between the observations and the fitted regression line.

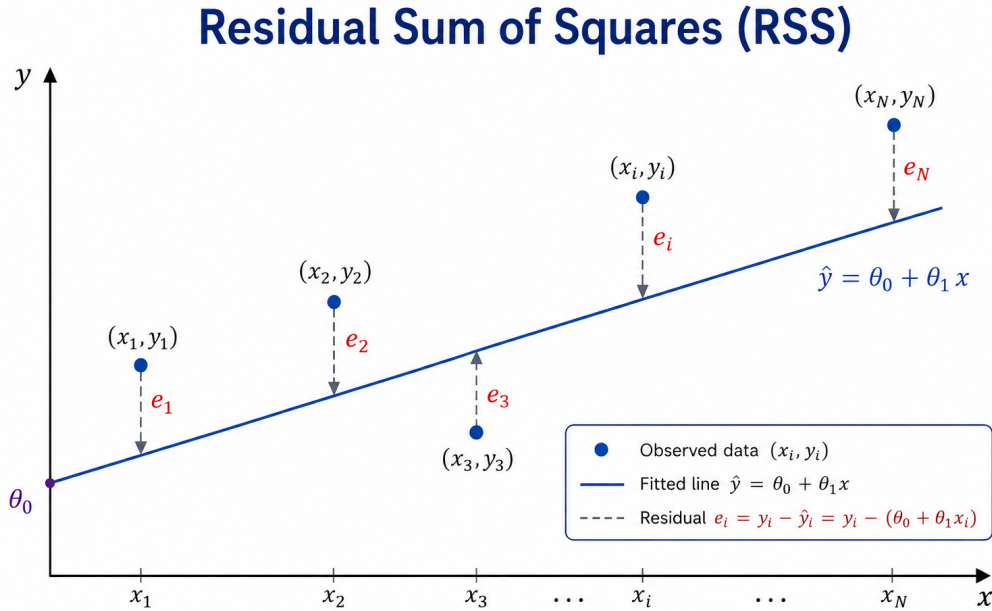


Figure 1.9: Residual Sum of Squares (RSS)

The ordinary least squares estimators are obtained by minimizing Equation 1.6 with respect to the model parameters. Since the RSS is a convex quadratic function of θ_0 and θ_1 , the solution can be obtained by setting the gradient equal to zero. The convexity of the RSS follows from the positive definiteness of the associated Hessian matrix, ensuring that the stationary point corresponds to a unique global minimum.

The resulting estimators are given by

$$\hat{\theta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)} \quad (1.7)$$

and

$$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \quad (1.8)$$

Equation 1.7 shows that the slope estimator can be interpreted as the ratio between the covariance of x and y and the variance of x . Consequently, the estimated slope increases when both variables tend to move together and decreases when their linear relationship weakens. In the context of empirical asset pricing, this interpretation leads naturally to the CAPM market beta, which can be viewed as the ordinary least squares estimator obtained by regressing the excess return of an asset against the excess return of the market portfolio. Under this interpretation, the CAPM beta measures the linear sensitivity of the asset return to market movements in the least-squares sense.

1.5.1.2 Residual Analysis and Model Assumptions

Once the regression coefficients have been estimated, the residuals provide information about the portion of the variability that is not explained by the model. For each observation, the residual is defined as

$$e_i = y_i - \hat{y}_i \quad (1.9)$$

where $\hat{y}_i$ denotes the predicted value obtained from the fitted regression model.

A distinction should be made between residuals and random errors. Residuals correspond to observable deviations between the fitted model and the data, whereas the random errors ε_i introduced in Equation 1.4 represent the unobservable stochastic component assumed by the statistical model.

Classical linear regression assumes that the random errors are independent, centered around zero, and have constant variance. Under these assumptions,

$$\varepsilon_i \sim N(0, \sigma^2) \quad (1.10)$$

where σ^2 denotes the variance of the error term.

The assumption of constant variance is commonly referred to as homoscedasticity and can be expressed as

$$\text{Var}(\varepsilon_i) = \sigma^2 \quad (1.11)$$

for all observations. When the variance of the errors changes across observations, the model is said to exhibit heteroscedasticity, which may affect the reliability of confidence intervals, hypothesis tests, and statistical inference.

Although these assumptions are rarely satisfied exactly in real financial data, they provide the statistical framework underlying ordinary least squares estimation and the interpretation of the inferential quantities discussed in the following sections.

1.5.1.3 Confidence Intervals and Standard Errors

The regression coefficients estimated through ordinary least squares are subject to sampling variability. Consequently, point estimates alone do not provide information about the uncertainty associated with the estimated parameters. This uncertainty is commonly quantified through standard errors and confidence intervals.

Under the assumptions introduced previously, the variance of the random error term can be estimated from the residuals through the residual standard error (RSE), defined as

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{N - 2}} \quad (1.12)$$

where N denotes the number of observations and the subtraction of two degrees of freedom reflects the estimation of the intercept and slope coefficients.

The standard errors associated with the regression coefficients are given by

$$\text{SE}(\hat{\theta}_1) = \sqrt{\frac{\sigma^2}{\sum_{i=1}^N (x_i - \bar{x})^2}} \quad (1.13)$$

and

$$\text{SE}(\hat{\theta}_0) = \sqrt{\sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \right]} \quad (1.14)$$

where σ^2 is typically replaced in practice by the estimator introduced in Equation 1.12.

These expressions show that the uncertainty associated with the estimated coefficients depends on both the residual variability and the dispersion of the predictor variable. In particular, the standard error of the slope coefficient decreases as the variability of x increases, leading to more stable estimates of the linear relationship.

Confidence intervals provide a range of plausible values for the regression coefficients. Under approximate normality assumptions, a 95% confidence interval for a parameter θ can be written as

$$\hat{\theta} \pm 2 \cdot \text{SE}(\hat{\theta}) \quad (1.15)$$

indicating the range within which the true parameter value is expected to lie with approximately 95% confidence.

1.5.1.4 Hypothesis Testing and Statistical Significance

Beyond estimating the regression coefficients, it is often important to evaluate whether the observed linear relationship is statistically significant. In simple linear regression, this is commonly assessed through hypothesis testing on the slope coefficient.

The null and alternative hypotheses are given by

$$\begin{cases} H_0 : \theta_1 = 0 \\ H_1 : \theta_1 \neq 0 \end{cases} \quad (1.16)$$

Under the null hypothesis, the predictor variable does not contribute to explaining the variability of the response variable, implying the absence of a linear relationship between x and y .

The statistical significance of the estimated slope coefficient can be evaluated through the t-statistic,

$$t = \frac{\hat{\theta}_1}{\text{SE}(\hat{\theta}_1)} \quad (1.17)$$

which measures the magnitude of the estimated coefficient relative to its standard error.

The corresponding p-value quantifies the probability of observing a statistic at least as extreme as the one obtained under the null hypothesis. Small p-values provide evidence against H_0 , suggesting that the predictor variable contributes significantly to the regression model.

In the context of the CAPM, hypothesis testing is particularly relevant for evaluating whether the estimated market exposure coefficient differs significantly from zero and whether the intercept term is statistically consistent with the theoretical prediction of $\alpha = 0$.

1.5.1.5 Analysis of Variance (ANOVA)

The variability of the response variable can be decomposed into a component explained by the regression model and a residual component associated with unexplained variability. In simple linear regression, this decomposition is expressed as

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i) \quad (1.18)$$

which separates the total deviation of each observation into an explained component and a residual component.

Figure 1.10 illustrates the geometric interpretation of this decomposition in the context of simple linear regression.

Summing over all observations leads to the classical ANOVA decomposition

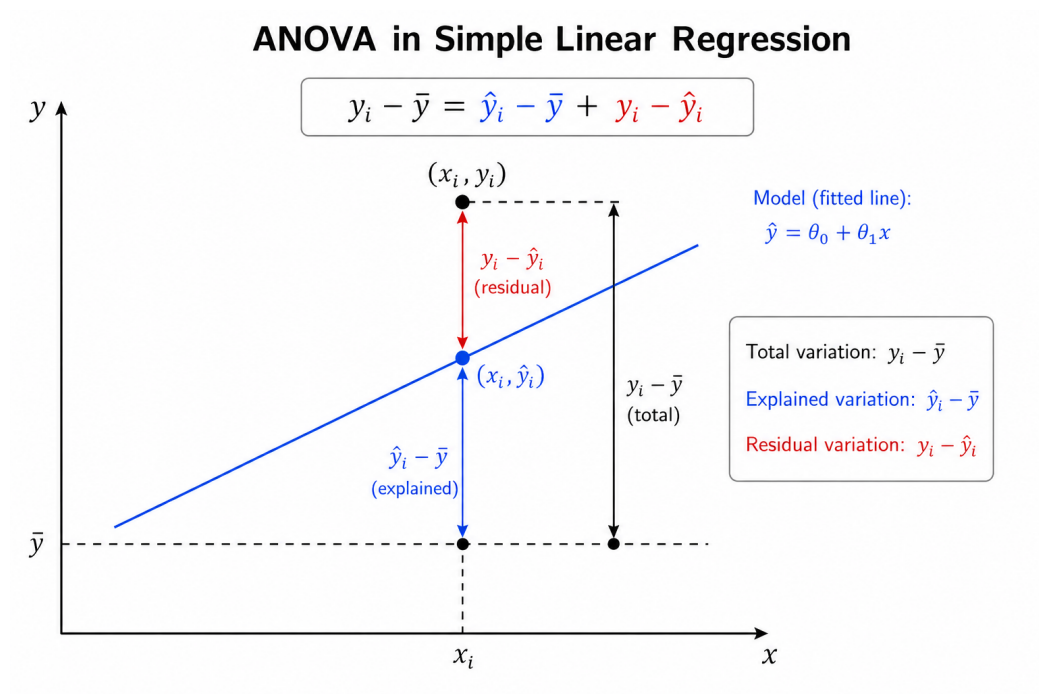


Figure 1.10: ANOVA decomposition in simple linear regression

$$\text{TSS} = \text{MSS} + \text{RSS} \quad (1.19)$$

where

$$\text{TSS} = \sum_{i=1}^N (y_i - \bar{y})^2 \quad (1.20)$$

represents the total variability of the response variable,

$$\text{MSS} = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2 \quad (1.21)$$

corresponds to the variability explained by the regression model, and

$$\text{RSS} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (1.22)$$

measures the unexplained residual variability.

Dividing each sum of squares by its associated degrees of freedom leads to the corresponding mean squares. In simple linear regression, the mean model square and the mean residual square are given by

$$\text{MMS} = \text{MSS} \quad (1.23)$$

and

$$\text{MSR} = \frac{\text{RSS}}{N - 2} \quad (1.24)$$

respectively.

These quantities allow the construction of the F-statistic,

$$F = \frac{\text{MMS}}{\text{MSR}} \quad (1.25)$$

which compares the variability explained by the regression model with the unexplained residual variability. Large values of the F-statistic suggest that the regression model explains a substantial fraction of the observed variability relative to the residual noise.

1.5.1.6 Coefficient of Determination (R^2)

Another commonly used measure in linear regression is the coefficient of determination, denoted by R^2 . This statistic quantifies the proportion of the total variability in the response variable that is explained by the regression model.

Using the ANOVA decomposition introduced in Equation 1.19,

$$\text{TSS} = \text{MSS} + \text{RSS}$$

the coefficient of determination can be written as

$$R^2 = \frac{\text{MSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \quad (1.26)$$

Larger values of R^2 indicate that the model explains a greater fraction of the observed variability in the data. In particular, $R^2 = 1$ corresponds to a perfect fit, whereas $R^2 = 0$ indicates that the regression model does not explain the variability of the response variable better than the sample mean alone.

In the special case of simple linear regression, the coefficient of determination is directly related to the Pearson correlation coefficient through

$$R^2 = r^2 \quad (1.27)$$

where r denotes the sample correlation between the predictor variable x and the response variable y .

Although R^2 provides a useful measure of goodness of fit, a large value does not necessarily imply that the model is statistically valid or economically meaningful. For this reason, R^2 is typically interpreted together with residual diagnostics, hypothesis tests, and domain-specific considerations.

1.5.2 Multiple Linear Regression and the Fama-French Model

While the CAPM can be formulated as a simple linear regression model with a single explanatory factor, the Fama–French three-factor model extends this framework by incorporating additional sources of systematic risk. From a statistical perspective, this leads naturally to a multiple linear regression model in which the response variable depends simultaneously on several predictors.

For a model with M explanatory variables, the linear regression model can be written as

$$y_i = \theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_M x_{iM} + \varepsilon_i \quad (1.28)$$

where θ_0 represents the intercept, $\theta_1, \dots, \theta_M$ denote the regression coefficients associated with the predictor variables, and ε_i corresponds to the random error associated with observation i .

Using matrix notation, the model can be expressed compactly as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon} \quad (1.29)$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1M} \\ 1 & x_{21} & \cdots & x_{2M} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{NM} \end{bmatrix}$$

and

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_M \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{bmatrix}$$

In this representation, the rows of $\mathbf{X}$ correspond to observations, while the columns represent the explanatory variables included in the model. The matrix

formulation provides a compact and computationally efficient framework for parameter estimation and statistical inference.

Under this interpretation, the Fama–French three-factor model can be viewed as a particular case of multiple linear regression in which the excess return of an asset is explained simultaneously by the market factor, the SMB factor, and the HML factor.

1.5.2.1 Ordinary Least Squares Estimation

As in the simple linear regression setting, the parameters of the multiple linear regression model are estimated by minimizing the residual sum of squares. Using the matrix formulation introduced in Equation 1.29, the residual vector can be written as

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\theta \quad (1.30)$$

and the corresponding residual sum of squares is given by

$$\text{RSS} = \mathbf{e}^T \mathbf{e} = (\mathbf{y} - \mathbf{X}\theta)^T (\mathbf{y} - \mathbf{X}\theta) \quad (1.31)$$

The ordinary least squares estimator is obtained by minimizing Equation 1.31 with respect to the parameter vector θ . As in the simple linear regression case, the optimization problem is convex and admits a unique global minimum provided that the columns of the design matrix $\mathbf{X}$ are linearly independent.

The resulting estimator is given by

$$\hat{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (1.32)$$

which generalizes the ordinary least squares estimator introduced previously for simple linear regression.

The matrix formulation highlights an important aspect of multiple linear regression: each coefficient measures the linear contribution of its associated predictor while accounting for the remaining variables included in the model. In the context of the Fama–French model, this means that the estimated coefficients

associated with the market, SMB, and HML factors represent their partial contributions to the excess return of the asset after controlling for the other factors included in the regression.

1.5.2.2 Statistical Interpretation of the Coefficients

In multiple linear regression, each regression coefficient measures the expected variation in the response variable associated with a one-unit change in the corresponding predictor, while holding the remaining predictors fixed. This interpretation differs from simple linear regression, where the effect of the predictor is evaluated without conditioning on additional explanatory variables.

Under the ordinary least squares framework, the uncertainty associated with the estimated parameter vector can be characterized through its covariance matrix, given by

$$\text{Cov}(\hat{\theta}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1} \quad (1.33)$$

where σ^2 denotes the variance of the random error term.

As in the simple linear regression setting, the variance of the residuals can be estimated through the residual standard error,

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{N - M - 1}} \quad (1.34)$$

where N represents the number of observations and M the number of explanatory variables included in the model.

The diagonal elements of Equation 1.33 determine the standard errors associated with each regression coefficient and provide a measure of the uncertainty of the parameter estimates. Larger standard errors indicate greater uncertainty in the estimated contribution of the corresponding predictor.

In the context of the Fama–French three-factor model, the estimated coefficients associated with the market, SMB, and HML factors measure the sensitivity of the asset return to each systematic risk factor after controlling for the remaining factors included in the regression. Consequently, the interpretation of each coefficient should always be understood conditionally on the presence of the other explanatory variables.

The inclusion of multiple predictors may also introduce statistical challenges such as multicollinearity, which occurs when explanatory variables exhibit strong linear dependence. In such cases, coefficient estimates may become unstable and difficult to interpret, even when the overall predictive performance of the model remains satisfactory.

1.5.2.3 Global Hypothesis Testing and Model Significance

In multiple linear regression, it is often important to evaluate whether the explanatory variables collectively contribute to the model. This can be assessed through a global hypothesis test of the form

$$\begin{cases} H_0 : \theta_1 = \theta_2 = \dots = \theta_M = 0 \\ H_1 : \text{at least one } \theta_j \neq 0 \end{cases} \quad (1.35)$$

Under the null hypothesis, none of the explanatory variables contribute to explaining the variability of the response variable, implying that the model does not perform better than using the sample mean alone.

The global significance of the regression model can be evaluated through the F-statistic,

$$F = \frac{(\text{TSS} - \text{RSS})/M}{\text{RSS}/(N - M - 1)} \quad (1.36)$$

which compares the variability explained by the model with the unexplained residual variability after adjusting for the corresponding degrees of freedom.

Large values of the F-statistic suggest that the explanatory variables collectively explain a substantial fraction of the observed variability in the response variable. The corresponding p-value provides a quantitative measure of the evidence against the null hypothesis.

In the context of the Fama–French three-factor model, the global F-test evaluates whether the market, SMB, and HML factors jointly contribute to explaining the excess return of the asset. Consequently, the test provides a global assessment of the explanatory power of the multifactor specification relative to a model without risk factors.

1.5.2.4 Limitations of Multiple Linear Regression

Although multiple linear regression provides a flexible and interpretable framework for modeling relationships between variables, its performance depends on several assumptions and modeling choices. In practice, violations of these assumptions may affect the reliability of parameter estimates, confidence intervals, and hypothesis tests.

One important limitation is multicollinearity, which arises when explanatory variables exhibit strong linear dependence. In such cases, the regression coefficients may become unstable and highly sensitive to small changes in the data, making their interpretation more difficult.

Multiple linear regression models may also be affected by omitted variable bias when relevant explanatory factors are excluded from the specification. In empirical asset pricing, this issue motivated the development of multifactor models such as the Fama–French framework, which extends the CAPM by incorporating additional sources of systematic risk beyond the market factor alone.

Another practical limitation is that linear regression assumes a linear relationship between predictors and the response variable. While this assumption often provides a useful approximation, real financial data may exhibit nonlinear effects, structural changes, heteroscedasticity, or temporal dependence that are not fully captured by the model.

Despite these limitations, linear regression models remain fundamental tools in empirical finance due to their interpretability, computational efficiency, and strong connection with statistical inference and risk factor modeling.

Chapter 2

Unsupervised Learning for Asset Allocation

Work in progress.

2.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

2.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

2.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

2.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

2.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

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Part II

Retail Analytics

Overview

This part covers problems related to retail analytics, including the analysis of customer behavior, transaction data, and marketing performance.

The scope of this domain includes methods for segmentation, prediction, and decision support, with applications in areas such as fraud detection, customer lifetime value estimation, and campaign optimization.

Chapter 3

Credit Card Fraud Detection

Work in progress.

3.1 Problem Definition

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Typical questions considered in this section include:

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Chapter 4

Customer Lifetime Value Prediction

Work in progress.

4.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

4.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

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4.3 Methodology and Implementation

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Typical topics may include:

- Data collection and sources
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Chapter 5

Marketing Campaign Optimization

Work in progress.

5.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

5.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

5.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

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- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

5.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

5.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

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Chapter 6

RFM Segmentation with K-Means

Work in progress.

6.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

6.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

6.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

6.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

6.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 7

Customer Sentiment and Topic Analysis

Work in progress.

7.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

7.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

7.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

7.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

7.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Part III

Biological Systems

Overview

This part covers problems related to biological systems, including the analysis and modeling of data arising in biological, health, and life science contexts.

The scope of this domain includes approaches for studying complex processes, identifying patterns in biological data, and supporting analysis in settings where systems are often heterogeneous, nonlinear, and influenced by multiple interacting factors.

Chapter 8

Microbial Population Dynamics (Lotka–Volterra Model)

Work in progress.

8.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

8.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

8.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

8.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

8.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Part IV

Energy and Industrial Systems

Overview

This part covers problems related to energy and industrial systems, including the analysis and modeling of production processes, physical assets, and operational data.

The scope of this domain includes approaches for optimization and efficient resource allocation, monitoring and analysis of system integrity, and the study of degradation processes such as corrosion, as well as data-driven methods to support decision-making in complex operational environments.

Chapter 9

Geochemical Fingerprint Deconvolution

Work in progress.

9.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

9.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

9.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

9.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

9.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 10

Pore–Throat Detection in Porous Media

Work in progress.

10.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

10.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

10.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

10.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

10.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 11

Optimal Hedging of Industrial Gas Demand

Work in progress.

11.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

11.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

11.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

11.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

11.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Chapter 12

Knowledge Extraction from Technical Documents with NLP and LLMs

Work in progress.

12.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

12.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

12.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

12.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

12.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Part V

Climate and Environmental Systems

Overview

This part covers problems related to climate and environmental systems, including the analysis and modeling of temporal and spatial data describing atmospheric and large-scale natural processes.

The scope of this domain includes approaches for time series analysis, forecasting, and the study of variability and uncertainty in systems influenced by natural dynamics, external drivers, and interactions across multiple scales.

Chapter 13

Climate Forecasting with Time Series

Work in progress.

13.1 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

13.2 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology

- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

13.3 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

13.4 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

13.5 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations
- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

Part VI

Writings & Essays

Overview

This part covers technical writings and essays spanning scientific modeling, data analysis, and related areas of mathematics, physics, and philosophy.

The scope includes discussions of modeling principles, reasoning under uncertainty, and computational approaches, providing context and deeper insight into the ideas underlying the projects.

Chapter 14

Data Science in the Age of Hype: Why the scientific method still matters

14.1 The Rise of Data Science

Data has become one of the most valuable assets of any modern organization. The unprecedented flow of information—resulting from countless digital interactions such as clicks, e-commerce transactions, mobile communications, and social media activity—reflects its central role in contemporary society. At the same time, the rapid advances in computing that enable large-scale data collection, storage, and processing are radically transforming the way businesses, governments, scientific communities, and institutions make decisions.

Yet, raw data by itself is meaningless. It only acquires value when subjected to systematic and rigorous analysis through methodologies capable of detecting patterns, inferring relationships, predicting trends, and ultimately translating insights into decisions with real-world impact. This is why Clive Humby’s now-famous analogy describes data as “the new oil”: like crude oil, data must be refined before it can generate value. The crucial element is not the “raw material” itself but the techniques used to extract its potential and the knowledge derived from it. This reflection naturally leads to a fundamental question: what do we really mean by Data Science?

Historically, the term Data Science emerged from an evolutionary process shaped by multiple contributions that helped formalize it as a discipline. Although its roots lie in statistics and computer science, its consolidation as an autonomous field took place over more than half a century of theoretical and technological developments.

The conceptual seed of Data Science can be traced back to 1962, when American statistician John W. Tukey published his influential essay *The Future of Data Analysis*. In it, Tukey argued that statistics should not be confined to hypothesis confirmation alone, but should place equal emphasis on the exploration, diagnosis, and understanding of data [15], [16]. He introduced *data analysis* as a distinct discipline, anticipating the vision of a “science of learning from data”—one in which empirical inquiry and iterative interaction with data occupy the core of quantitative knowledge.

Later, in 1974, Danish computer scientist Peter Naur, in his *Concise Survey of Computer Methods*, explicitly used the terms *Data Science* and *Datalogy* to refer to the systematic study of data [17]. Naur argued that computing should focus on the structure, meaning, and organization of data rather than on hardware or programming. Although his proposal went largely unnoticed for years, it laid conceptual foundations that would be revisited decades later.

In 1997, statistician C.F. Jeff Wu, during his inaugural lecture at the University of Michigan titled *Statistics = Data Science?*, proposed renaming statistics as *Data Science* and calling modern statisticians *data scientists* [18]. His goal was to signal a new era in which statistical inference should integrate with computation and empirical analysis to move beyond mere data description and outdated stereotypes.

This perspective was further developed in 2001 by William S. Cleveland in his seminal article *Data Science: an Action Plan for Expanding the Technical Areas of the Field of Statistics* [19]. Cleveland proposed a systematic expansion of statistics by identifying six core technical areas of Data Science: multidisciplinary investigations, models and methods for data, computing with data, pedagogy, tool evaluation, and theory. Rather than allowing statistics to contract into a narrowly mathematical discipline, he emphasized the central role of computation, real-world data analysis, education, and the development and evaluation of tools. His framework laid the conceptual groundwork for understanding Data Science as a comprehensive, data-centered field—deeply rooted in statistics, yet necessarily broader in scope and practice.

That same year, Leo Breiman, a statistician at the University of California, Berkeley, published his influential essay *Statistical Modeling: The Two Cultures*, in which he argued that statistical modeling serves two fundamental purposes: **information**—extracting insight into how nature associates response variables with input variables—and **prediction**—the ability to accurately anticipate future responses for new observations [20]. Breiman critically observed that traditional statistical practice had largely privileged the informational goal through highly structured, assumption-driven models, often at the expense of predictive performance. He advocated for a broader methodological perspective in which algorithmic and data-driven approaches play a central role, emphasizing prediction as an essential complement rather than a secondary objective. This distinction, and the tension it revealed between explanation and prediction, foreshadowed core principles of modern Data Science, where understanding and predictive accuracy are treated as intertwined but distinct aims.

By the late 2000s, the rise of Big Data and the growing need for professionals capable of combining statistics, programming, and domain expertise fueled the popularization of both the term Data Science and the role of the data scientist. Media consolidation came in 2012, when Thomas H. Davenport and DJ Patil published *Data Scientist: The Sexiest Job of the 21st Century* in the *Harvard Business Review* [21]. Since then, Data Science has become a strategic discipline across both academia and industry.

Today, Data Science is recognized as an interdisciplinary field that integrates statistical, computational, and conceptual methods to extract knowledge, infer patterns, and generate predictions from data. In many ways, it represents the natural evolution of statistics a discipline now equipped to address complex phenomena in fields as diverse as economics, biology, marketing, and physics.

However, despite its relevance and expansion, the term *Data Science* has become somewhat diffuse, often distorted by media overexposure. This ambiguity calls for a deeper examination of its epistemological foundations: if we speak of a **Science** of Data, we must ask what makes a discipline a science?

From this perspective, Data Science can indeed be considered a science. Its scientific character lies in the systematic application of the scientific method to the study of data. The data scientist formulates hypotheses, analyzes observations, and tests results just as any other scientist does. The difference lies in the object of study: data as a quantitative representation of reality.

In this sense, Data Science can be viewed as a generalization of statistics extending its focus from **inference** (understanding how the world behaves based on observed data) to **prediction** (anticipating how it might behave under new conditions). This predictive capability, powered by machine learning algorithms, enables the effective application of the **Hypothetical-Deductive Method**: formulating hypotheses about future phenomena and testing them through data and predictive models. In other words, machine learning acts as a technological enabler that materializes Breiman's inference–prediction duality, providing tools to model complex relationships directly from data and to test hypotheses that were previously unreachable with traditional statistical methods. It is this integration of inference, prediction, and empirical validation that solidifies the scientific nature of the discipline.

Understanding these foundations is essential for interpreting the true scope of Data Science. While the discipline draws from statistics, computer science, and domain expertise, its ultimate objective is not merely the manipulation of data but the extraction of reliable knowledge about the world. In this sense, Data Science should not be understood as a collection of algorithms or software tools, but as a methodological framework for reasoning under uncertainty.

However, the rapid growth of the field has also brought an unintended consequence: a proliferation of narratives that portray Data Science primarily as a technological revolution driven by ever more sophisticated algorithms. In many contexts, the emphasis has shifted toward tools, automation, and predictive performance, sometimes overshadowing the epistemological principles that historically guided scientific inquiry.

This tension raises a critical question for modern practitioners: if Data Science claims to be a science, what role does the scientific method play in its practice today? Addressing this question requires revisiting the principles that define scientific reasoning and examining how they apply to the analysis and interpretation of data in an era increasingly shaped by computational power and algorithmic models.

14.2 The Nature of Science and the Scientific Method

The very semantics of the term *Data Science* make it impossible to detach it from the concept of *science*. This forces us to pause and reflect: can Data Science truly

be regarded as a scientific discipline, or is it merely a fashionable commercial label? A comprehensive analysis of the nature of science lies far beyond the scope of this article; however, for the sake of intellectual honesty, we should not evade the issue. At the very least, we ought to clarify what makes an activity genuinely *scientific*.

From an epistemological perspective, science cannot be defined merely as any endeavor that leads to new discoveries or expands knowledge. Scientific knowledge evolves what was once considered true has often been revised, refined, or discarded. It is therefore reasonable to assume that part of what we regard today as scientifically valid will eventually be replaced, even if we cannot foresee which part. In other words, science does not guarantee truth. What distinguishes science from other forms of inquiry is not the *results* it produces, but the *method* it employs to generate them [22]. Science is better understood as an iterative mechanism—almost algorithmic in nature—that produces, tests, refines, and occasionally rejects knowledge. This is a central idea, because in essence, a data scientist does exactly that: applies the scientific method as a repeatable process of inquiry. We will return to this point shortly when discussing the *Hypothetico-Deductive Method*.

With this foundation in place, it becomes easier to see why Data Science can be viewed as a generalization of Statistics. This is not a trivial claim. Statistics is undoubtedly a science specifically, a formal science, alongside logic and computer science. Formal sciences study abstract systems through reasoning rather than direct experimentation, distinguishing them from natural, social, and applied sciences. Yet classical statistics has been historically grounded in *inductive inference*: reasoning from the particular to the general. Induction allows us to infer broader conclusions from observed cases, but it carries a fundamental limitation: inductive reasoning can *support* a conclusion, but never *prove* it. This problem, famously articulated by David Hume in the 18th century and known as the *problem of induction*, arises from the logical gap between past observations and future expectations.

Bertrand Russell illustrated this issue through the well-known “inductivist turkey”. A turkey observes that every morning at 9 a.m. the farmer brings food. After hundreds of consistent observations—rainy days, sunny days, warm and cold alike—the turkey confidently concludes that it will always be fed at 9 a.m. Yet on Christmas Eve, instead of being fed, it is slaughtered. A single contradictory event collapses a conclusion seemingly supported by

overwhelming evidence. The lesson is clear: no amount of past observations can logically ensure the truth of a universal claim about the future. Karl Popper pushed this idea further: induction can never confirm a theory with certainty, whereas a single counterexample can refute it. The turkey’s mistake—rooted in the “turkey illusion”—was assuming regularity without understanding the underlying mechanism (the farmer’s intention).

None of this implies that induction is a “bad” method. It remains essential as the generator of hypotheses and conjectures. As Klimovsky notes [23], induction may be seen as a probabilistic justification: it evaluates how strongly observations support a hypothesis, rather than certifying its truth. Put simply, science advances by combining induction (to propose hypotheses) and deduction (to test them). Deduction enables us to assess whether predictions derived from hypotheses correspond to reality. Verification, however, does not grant absolute truth; it merely confirms that a hypothesis has survived a test under certain conditions.

This tension lies at the heart of Leo Breiman’s critique of classical statistics. While statistics provided the mathematical language to quantify uncertainty and draw inferences about populations from limited data, it remained fundamentally bound to inductive reasoning: no matter how large the sample or how robust the model, a future observation could always contradict it. Machine Learning transformed this landscape by not only inferring patterns from data but also validating predictive performance empirically. A model is no longer judged solely by how well it explains past data, but by how accurately it predicts *unseen* data. This represents a qualitative shift that operationalizes the hypothetico-deductive paradigm: generating hypotheses (models) and testing them empirically against new observations in a reproducible and quantifiable way.

To illustrate this in a Data Science context, consider a credit-scoring model. Classical statistics may estimate relationships between customer features and default risk through inductive inference on historical samples. A machine learning model, however, must *demonstrate its predictive power* on new customers those the model has never encountered. If the predictions fail, the model is rejected or refined. This is falsification in action, applied to data-driven decision-making.

In this sense, Data Science can be expressed as **Inference + Prediction**, or equivalently, as **Statistics + Machine Learning**. It unites the inductive strength of statistics with the deductive rigor of model testing, fulfilling Breiman’s vision of a discipline that bridges explanation and prediction a modern science of learning from data. The next section will examine this duality through the lens of

the *Hypothetico-Deductive Method*, the framework that ultimately solidifies the scientific character of Data Science.

14.3 The Scientific Mindset of a Data Scientist

At the core of scientific inquiry lies a logical structure that transcends individual disciplines: the **Hypothetico-Deductive Method**. It provides the conceptual backbone of the scientific method, linking theoretical conjecture with empirical observation. The process begins with a hypothesis—a tentative model proposed to explain observed behavior or predict its future course. From this hypothesis, one deduces empirically testable consequences. These predictions are then confronted with real data through experimentation or systematic observation. When evidence contradicts the hypothesis, it must be revised or discarded; when the evidence supports it, the hypothesis gains provisional credibility, though never definitive truth.

This dynamic interplay between theoretical formulation, deduction, and empirical scrutiny defines the essence of scientific work. What grants scientific legitimacy to a model is not its mathematical elegance or computational sophistication, but its **capacity to generate falsifiable predictions**. The scientific method is therefore inherently self-correcting: each cycle of hypothesizing, testing, and refining moves knowledge closer to truth while acknowledging that truth remains an ever-distant horizon.

Traditional inferential statistics implements this logic only partially, primarily through hypothesis testing. A null hypothesis is proposed, its probabilistic implications are derived, and sample data are used to evaluate whether they justify its rejection. Yet this remains predominantly an inductive process, generalizing from finite observations into probabilistic claims about broader populations. Moreover, statistical validation is typically static once the data are gathered, the hypothesis is evaluated solely in that context, rather than through iterative confrontation with new evidence.

Machine learning, by contrast, operationalizes the deductive stage of scientific reasoning through **predictive validation**. Each model can be viewed as a hypothesis about the functional relationship between predictors and outcomes. Training a model corresponds to formulating the hypothesis; evaluating it on unseen data corresponds to testing its empirical consequences. Performance metrics

quantify how well the model withstands empirical scrutiny, and failures become informative, guiding the reformulation or replacement of the underlying hypothesis.

This perspective turns Data Science into a **practical enactment of the Hypothetico-Deductive Method**. It combines inductive discovery of structure within data with deductive testing of the predictive consequences of that structure. More importantly, it enables Data Science to progress from mere correlation to **genuine explanation**, seeking models that capture mechanisms rather than patterns alone.

Consider the problem of **customer churn** in retail. A predictive model might accurately identify customers at risk of leaving based solely on behavioral correlations a purely inductive exercise. But if the goal is to determine whether a discount campaign will *reduce* churn, prediction alone is insufficient. The data scientist must investigate—and, when possible, formalize—**causal models** capturing how interventions alter outcomes. Prediction answers *who* is likely to churn; causality answers *why* and *what we can do about it*.

Under the Hypothetico-Deductive paradigm, this iterative cycle can be summarized visually, as shown in Figure 14.1, and is typically composed of the following stages:

- **Systematic observation** of a phenomenon
- **Formulation of hypotheses** or models to explain observed patterns
- **Deduction of testable predictions** grounded in those hypotheses
- **Empirical evaluation** of predictions via new data
- **Revision or replacement** of hypotheses based on the evidence collected

Viewed from this angle, the role of a data scientist extends far beyond executing algorithms or statistical routines. It is the **iterative application of the scientific method** across diverse business contexts generating hypotheses, building predictive and causal models, confronting them with evidence, and allowing data to arbitrate which explanations survive. Data Science thus emerges not merely as a technical discipline, but as the **contemporary expression of scientific thinking** applied to data-rich environments, where distinguishing correlation from causation and prediction from explanation defines true professional excellence.

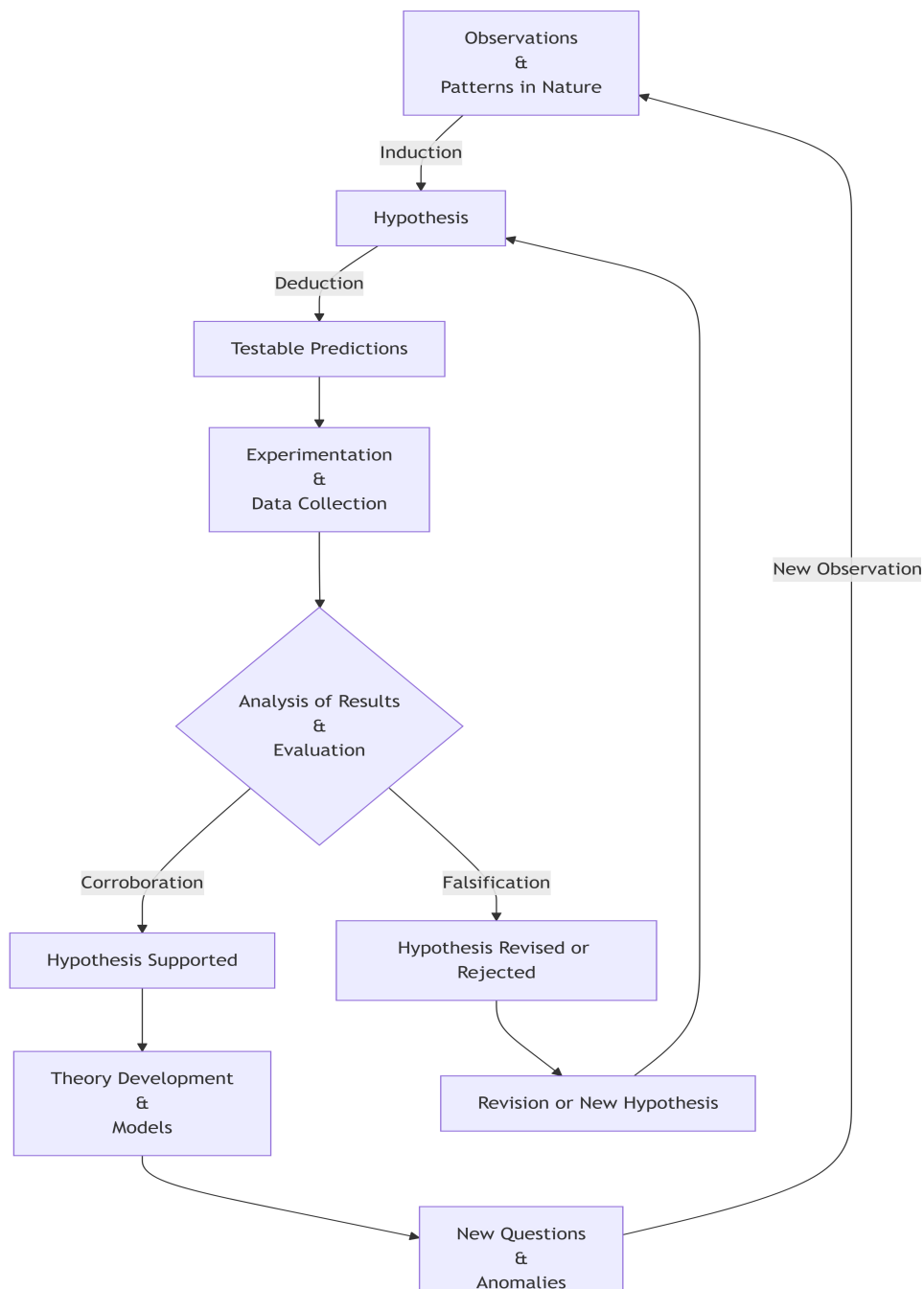


Figure 14.1: The hypothetico-deductive method as an iterative loop: observations motivate hypotheses (induction); hypotheses yield testable predictions (deduction); empirical results either support or challenge the hypothesis, leading to theory formation or revision and ultimately to new questions that restart the process.

14.4 The Fundamental Constituents of Data Science

Despite its rapid expansion and increasing visibility, Data Science is often portrayed as a discipline defined by an ever-growing collection of tools, frameworks, and technological trends. New libraries, platforms, and modeling techniques appear constantly, creating the impression that the field evolves primarily through technological novelty. This perception, however, can be misleading. While modern practice indeed relies on a rich computational ecosystem, the scientific core of Data Science is not defined by software stacks or fashionable methodologies. Rather, it rests on a far more compact and principled foundation: **probabilistic reasoning and statistical inference**.

From a first-principles perspective, Data Science is best understood as a systematic framework for reasoning under uncertainty using data. Probability provides the language to model randomness and variability, while Statistics supplies the methodological machinery for learning from data and evaluating the reliability of the conclusions we draw from it. Together, they form the conceptual backbone that allows data-driven reasoning to remain coherent, interpretable, and scientifically grounded.

Computation plays a crucial—but often misunderstood—role in this framework. The term *computation* derives from the Latin *computatio*, formed by the prefix *com* (“together” or “with”) and *putatio*, related to *putare*, meaning “to calculate” or “to reckon.” In its original sense, the word conveys the idea of **calculating together**. In this light, computation can be understood not as an autonomous source of knowledge, but as a process through which machines assist human reasoning by carrying out calculations at scales and speeds beyond our natural capacity. Algorithms, programming languages, and computational environments therefore act as instruments that enable probabilistic models to be simulated, statistical procedures to be implemented, and empirical hypotheses to be tested. Computation does not define Data Science; it provides the **operational medium** through which probabilistic and statistical reasoning becomes executable in practice.

14.5 Data Science as a Strategic Value Engine

At this point, it is crucial to emphasize a key principle that often gets overlooked: **Data Science is not a goal in itself**. No matter how sophisticated the methodolo-

gies or how advanced the machine learning models, they are worthless if they do not generate real and measurable impact. The true purpose of Data Science is to inform and improve strategic decisions—transforming data into outcomes that advance organizational goals.

Data-driven decision making empowers organizations to optimize operations, personalize customer experiences, identify risks proactively, and maintain a competitive advantage. As Kozyrkov highlights, outstanding data analysts ensure that their work is focused on solving the **right business problems**, not simply producing technically impressive outputs [24].

This underscores a critical truth: technical excellence without business alignment leads to solutions that may appear successful internally but **fail to influence decisions or create value**. The impact of a model is not measured solely by statistical accuracy or computational ingenuity, but by the extent to which it **changes what the organization does**.

From a decision-theoretic perspective, integrating Data Science into strategy enables measurable improvements not only in profitability or utility, but also in **decision velocity**. Automated analytical pipelines allow organizations to deploy models directly into operational environments continuously monitoring performance, adapting to new data, and discarding approaches that no longer deliver value. This creates a self-correcting loop where models are evaluated and refined over time, accelerating the pathway from insight to action.

Consider, for example, the case of **customer churn** in retail. Without Data Science, retention strategies typically rely on intuition or broad marketing campaigns, often costly and ineffective. In contrast, a churn prediction model can **identify customers most at risk**, enabling targeted interventions such as personalized offers or loyalty programs. If the strategy does not reduce churn among the targeted segment, performance monitoring mechanisms reveal the failure, prompting refinement or alternative plans. Here, data does not merely describe behavior—it actively shapes business outcomes.

Viewed through this lens, Data Science becomes a **strategic value engine**: a discipline that leverages scientific reasoning, computational tools, and domain expertise to drive decisions that matter. This synthesis—model-driven insight aligned with real-world objectives—ensures that Data Science fulfills its true mission: transforming information into impact.

14.6 Why the Scientific Method Still Matters

The strategic impact of Data Science ultimately stems from something deeper than predictive models or analytical pipelines. Its real strength lies in the intellectual framework that guides how data is interpreted, hypotheses are formulated, and decisions are evaluated. The ability of Data Science to generate meaningful impact in organizations is therefore inseparable from the scientific principles that structure its practice.

In an era defined by rapid technological change and the constant emergence of new tools, it is tempting to define Data Science by its most visible artifacts: programming languages, machine learning libraries, cloud platforms, and ever-evolving frameworks. Yet these elements, while powerful, are ultimately transient. Technologies evolve, tools are replaced, and methodologies shift as the field continues to mature.

What endures is something far more fundamental: the scientific mindset that underlies the discipline. At its core, Data Science is not about algorithms but about **reasoning under uncertainty**, formulating hypotheses, confronting them with data, and refining our understanding of complex phenomena through systematic evidence. Probability provides the language for uncertainty, statistics supplies the methods for inference, and computation serves as the medium through which these ideas become operational.

When practiced in this way, Data Science becomes more than a collection of analytical techniques—it becomes a modern expression of the scientific method applied to data-rich environments. Its true power lies not in automation or predictive accuracy alone, but in its capacity to transform data into knowledge and knowledge into better decisions. In the end, the enduring value of Data Science does not come from the sophistication of its tools, but from the rigor of the thinking that guides their use.

The original version is available on Medium [here](#).

Chapter 15

Conformal Geometry and Electrostatics in Three Dimensions

15.1 Laplace's equation

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (15.1)$$

15.2 Conformal Mappings

15.3 Problem Definition

This section describes the problem being addressed and explains its relevance within the context of the domain.

Typical questions considered in this section include:

- What is the objective of the project?
- Why is the problem relevant?
- What type of decisions or insights depend on the solution?
- What are the main challenges associated with the problem?

The emphasis is placed on the practical motivation and the broader context rather than on technical implementation details.

15.4 Modeling Approach

This section introduces the conceptual framework used to approach the problem.

The discussion typically includes:

- The main modeling assumptions
- The reasoning behind the selected methodology
- The intuition behind the model or algorithm
- The relationship between the model and the underlying process being studied

Mathematical expressions may be introduced when they help clarify the modeling decisions or the structure of the method. However, long derivations are generally avoided in the main text.

The emphasis is placed on communicating the logic of the approach rather than on developing a full mathematical treatment.

15.5 Methodology and Implementation

This section describes the data, experimental design, computational workflow, and implementation details used throughout the project.

Typical topics may include:

- Data collection and sources
- Feature engineering
- Cleaning and preprocessing
- Simulation strategies
- Training and validation methodology
- Experimental setup
- Numerical methods and algorithms
- Workflow organization
- Pipeline structure
- Computational considerations

The goal is to provide enough detail to understand how the analysis was conducted and implemented while maintaining a clear and readable narrative.

Implementation details are intended to complement the conceptual discussion without turning the chapter into a software manual.

References to notebooks and scripts are included throughout the project when relevant to the implementation or analysis.

15.6 Results, Discussion, and Limitations

This section presents the main outcomes of the project together with an interpretation of the results, model behavior, and practical limitations of the approach.

Typical elements may include:

- Visualizations and plots
- Performance metrics
- Comparisons with baseline approaches
- Sensitivity analysis
- Error analysis
- Interpretation of results
- Model assumptions and their consequences
- Sources of uncertainty
- Robustness and generalization
- Trade-offs associated with the methodology
- Potential future improvements

The emphasis is placed not only on performance, but also on understanding the behavior, limitations, and practical implications of the proposed solution.

15.7 Technical Appendix (Optional)

This optional section contains technical details that support the project but are not essential for following the main narrative.

Possible contents include:

- Mathematical derivations
- Additional proofs
- Numerical details
- Extended equations

- Supplementary experiments
- Technical notes on auxiliary methods and algorithms

The appendix may also include concise discussions of methodological concepts when they help clarify modeling decisions or support the interpretation of the results.

The main body of the project is intended to remain fully understandable without requiring consultation of the appendix.

The appendix serves as an optional technical reference for readers interested in deeper mathematical or computational details.

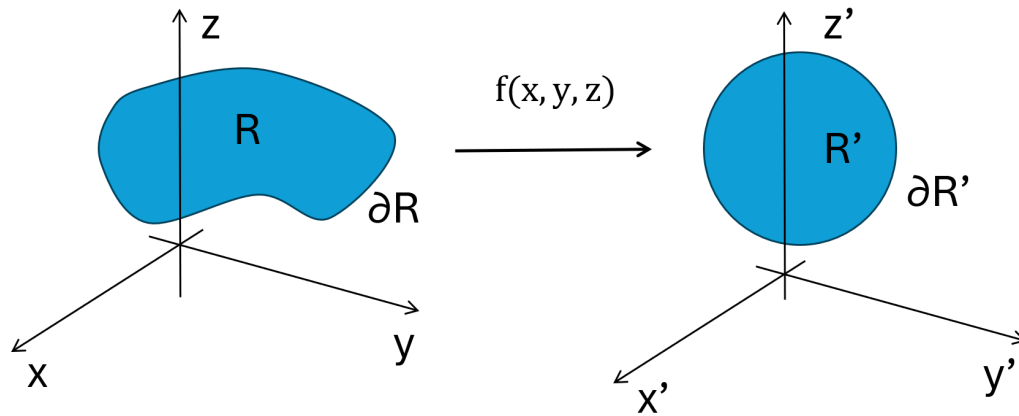


Figure 15.1: Schematic representation of a conformal transformation $f(x, y, z)$ mapping a region R with boundary ∂R in the (x, y, z) coordinate system into a transformed region R' with boundary $\partial R'$ in the (x', y', z') coordinate system.

As shown in Figure 15.1, ...

Suppose we have a region R in xyz space in which a scalar field ϕ satisfies Laplace's equation Equation 15.1. We want to determine what conditions a transformation $f : R \rightarrow R'$ must satisfy such that Laplace's equation remains invariant under this transformation. For simplicity, let's consider the simplest case in $\mathbb{R}^2$.

Applying the chain rule we have that:

$$\begin{aligned}\frac{\partial \phi}{\partial x} &= \frac{\partial \phi}{\partial x'} \frac{\partial x'}{\partial x} + \frac{\partial \phi}{\partial y'} \frac{\partial y'}{\partial x} \\ \frac{\partial \phi}{\partial y} &= \frac{\partial \phi}{\partial x'} \frac{\partial x'}{\partial y} + \frac{\partial \phi}{\partial y'} \frac{\partial y'}{\partial y}\end{aligned}\tag{15.2}$$

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